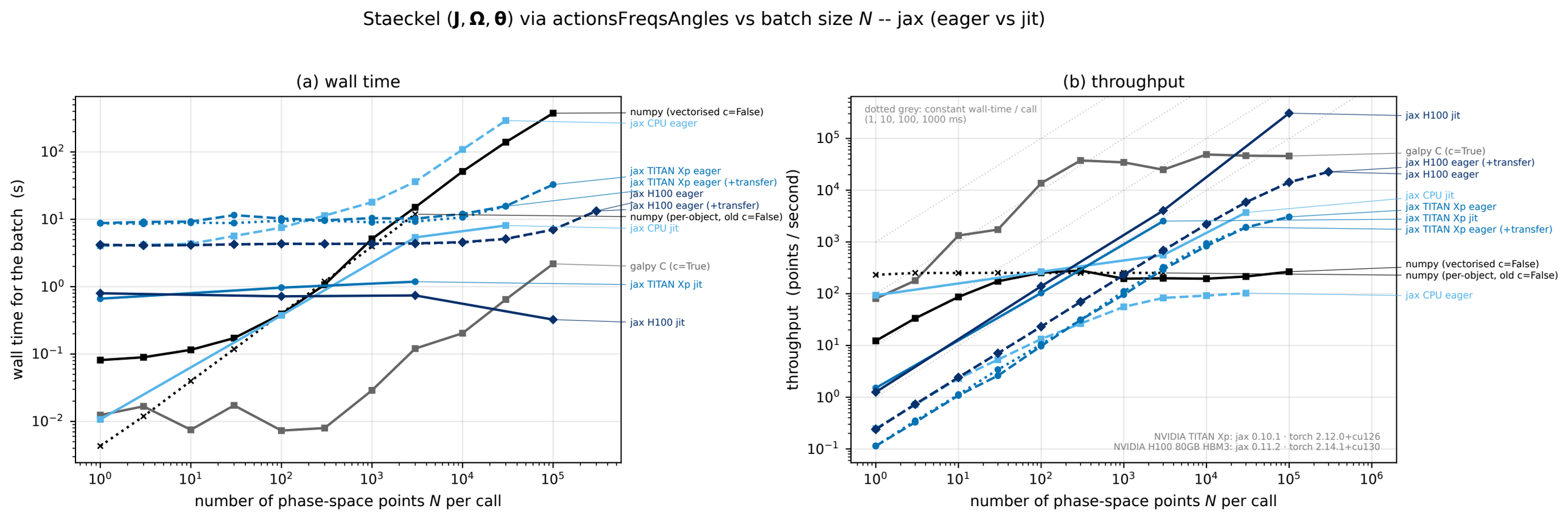
Staeckel (J, Ω, θ) via actionsFreqsAngles vs batch size N -- jax (eager vs jit)
(a) wall time
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised c=False)
jax CPU eager
jax TITAN Xp eager
jax TITAN Xp eager (+transfer)
jax H100 eager
jax H100 eager (+transfer)
numpy (per-object, old c=False)
jax CPU jit
galpy C (c=True)
jax TITAN Xp jit
jax H100 jit
(b) throughput
throughput (points / second)
number of phase-space points N per call
dotted grey: constant wall-time / call
(1, 10, 100, 1000 ms)
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130
jax H100 jit
galpy C (c=True)
jax H100 eager (+transfer)
jax H100 eager
jax CPU jit
jax TITAN Xp eager
jax TITAN Xp jit
jax TITAN Xp eager (+transfer)
numpy (vectorised c=False)
numpy (per-object, old c=False)
jax CPU eager